

CPS188: Lab 03 - Selection Control Structures

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Problem #1:

Calculate the user's training heart rate.

Algorithm:

1. Ask the user their gender, age, resting heart rate, and fitness level
2. Use the information the user inputted to calculate their maximum heart rate using the formula:
Males: $MHR = \frac{203.7}{1 + e^{0.033(age - 104.3)}}$
Females: $MHR = \frac{190.2}{1 + e^{0.0453(age - 107.5)}}$
3. Calculate the training heart rate using the information the user inputted and the maximum heart rate calculated previously, using the formula: $THR = (MHR - RHR) \times INTEN + RHR$
4. Display the training her rate

Code:

```
#include <stdio.h>
#include <math.h>
#define E 2.718281828459045235360 //euler's number

int main (void){
    //declaring variables:
    char gender, level;
    double age, rhr, level_value, mhr, thr;

    //gender:
    do{
        printf("Enter your gender (male 'M' or female 'F'): ");
        scanf(" %c", &gender);
    } while (gender!='M' && gender!='F');

    //age:
    do{
        printf("Enter your age: ");
        scanf("%lf", &age);
    } while(age<=0.0);
```

```

//resting heart rate (RHR):
do{
    printf("Enter your resting heart rate number: ");
    scanf("%lf", &rhr);
} while (rhr<=0);

//fitness level:
do{
    printf("Enter your fitness level (low 'L', medium 'M', or high
'H'): ");
    scanf(" %c", &level);
} while(level!='L' && level!='M' && level!='H');

//setting a double value for the fitness level:
if(level=='L'){
    level_value=0.55;
}
else if (level=='M'){
    level_value=0.65;
}
else{
    level_value=0.8;
}

//calculate the maximum heart rate:
if(gender=='M'){
    mhr=203.7/(1+pow(E,0.033*(age-104.3)));
}
else{
    mhr=190.2/(1+pow(E,0.0453*(age-107.5)));
}

//calculate THR:
thr=(mhr-rhr)*(level_value)+rhr;
printf("The training heart rate is: %1.1f", thr); //output the
training heart rate. %1.0lf --> would output the training heart rate
rounded to the nearest integer value

```

```
    return (0);  
}
```

Output:

Part (a):

Enter your gender (male 'M' or female 'F'): M

Enter your age: 19

Enter your resting heart rate number: 64

Enter your fitness level (low 'L', medium 'M', or high 'H'): M

The training heart rate is: 147

Part (b):

Enter your gender (male 'M' or female 'F'): F

Enter your age: 20

Enter your resting heart rate number: 63

Enter your fitness level (low 'L', medium 'M', or high 'H'): H

The training heart rate is: 162

Part (c):

Enter your gender (male 'M' or female 'F'): M

Enter your age: 63

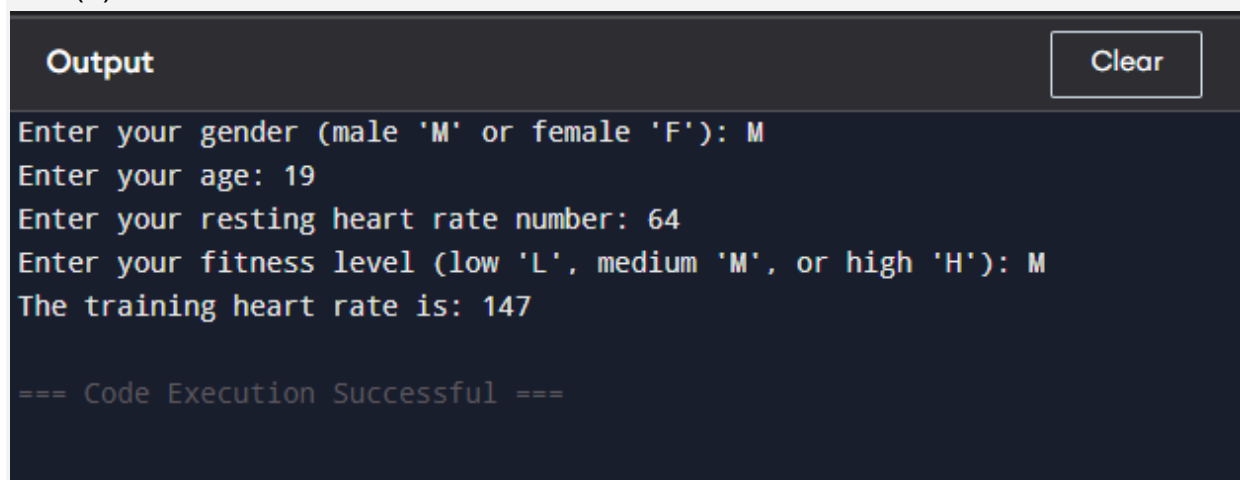
Enter your resting heart rate number: 82

Enter your fitness level (low 'L', medium 'M', or high 'H'): L

The training heart rate is: 126

Screenshot:

Part (a):



The screenshot shows a dark-themed output window. At the top right is a 'Clear' button. The output text is as follows:

```
Enter your gender (male 'M' or female 'F'): M  
Enter your age: 19  
Enter your resting heart rate number: 64  
Enter your fitness level (low 'L', medium 'M', or high 'H'): M  
The training heart rate is: 147  
  
=== Code Execution Successful ===
```

Part (b):

```
Output Clear

Enter your gender (male 'M' or female 'F'): F
Enter your age: 20
Enter your resting heart rate number: 63
Enter your fitness level (low 'L', medium 'M', or high 'H'): H
The training heart rate is: 162

=== Code Execution Successful ===|
```

Part (c):

```
Output Clear

Enter your gender (male 'M' or female 'F'): M
Enter your age: 63
Enter your resting heart rate number: 82
Enter your fitness level (low 'L', medium 'M', or high 'H'): L
The training heart rate is: 126

=== Code Execution Successful ===|
```

Problem #2:

Calculate and output the BMI of the user and accordingly output their classification.

Algorithm:

1. Ask the user to input their weight (in kilograms) and height (in meters)
2. Calculate the user's BMI using the following formula: $BMI = \frac{kg}{m^2}$
3. Output their BMI rounded to one decimal place and output their classification according to their BMI

Code:

```
#include <stdio.h>
#include <math.h>

int main (void){
    double weight, height, BMI;
```

```

    printf("Disclaimer: BMI is not used for muscle builders, long
distance athletes, pregnant women, the elderly or young
children.\n\n");
    do{
        printf("Enter your weight (in kilograms): ");
        scanf("%lf", &weight);
    } while(weight<=0);

    do{
        printf("Enter your height (in meters): ");
        scanf("%lf", &height);
    } while(height<=0);

    BMI=weight/(pow(height,2));

    //outputting according to BMI results:
    if(BMI<18.5){
        printf("Your BMI is %0.1lf and your classification is
underweight.", BMI);
    }
    else if(18.5<=BMI<=24.9){
        printf("Your BMI is %0.1lf and your classification is
normal.", BMI);
    }
    else if(25<=BMI<=29.9){
        printf("Your BMI is %0.1lf and your classification is
overweight.", BMI);
    }
    else{
        if(BMI>=30){
            printf("Your BMI is %0.1lf and your classification is
obese.", BMI);
        }
    }
    return (0);
}

```

Output:

Part (a):

Disclaimer: BMI is not used for muscle builders, long distance athletes, pregnant women, the elderly or young children.

Enter your weight (in kilograms): 81.5

Enter your height (in meters): 1.88

Your BMI is 23.1 and your classification is normal.

Part (b):

Disclaimer: BMI is not used for muscle builders, long distance athletes, pregnant women, the elderly or young children.

Enter your weight (in kilograms): 68

Enter your height (in meters): 1.55

Your BMI is 28.3 and your classification is normal.

Part (c):

Disclaimer: BMI is not used for muscle builders, long distance athletes, pregnant women, the elderly or young children.

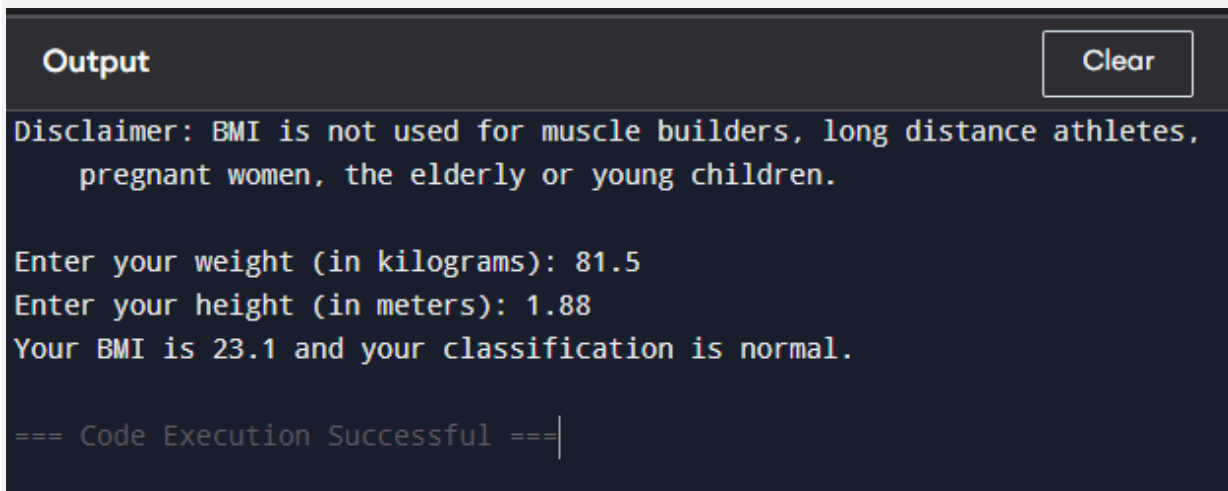
Enter your weight (in kilograms): 94

Enter your height (in meters): 1.57

Your BMI is 38.1 and your classification is normal.

Screenshot:

Part (a):



```
Output Clear  
Disclaimer: BMI is not used for muscle builders, long distance athletes,  
pregnant women, the elderly or young children.  
  
Enter your weight (in kilograms): 81.5  
Enter your height (in meters): 1.88  
Your BMI is 23.1 and your classification is normal.  
  
=== Code Execution Successful ===|
```

Part (b):

```
Output Clear

Disclaimer: BMI is not used for muscle builders, long distance athletes,
pregnant women, the elderly or young children.

Enter your weight (in kilograms): 68
Enter your height (in meters): 1.55
Your BMI is 28.3 and your classification is normal.

=== Code Execution Successful ===|
```

Part (c):

```
Output Clear

Disclaimer: BMI is not used for muscle builders, long distance athletes,
pregnant women, the elderly or young children.

Enter your weight (in kilograms): 94
Enter your height (in meters): 1.57
Your BMI is 38.1 and your classification is normal.

=== Code Execution Successful ===|
```

Problem #3:

Calculate the user's final grade.

Algorithm:

1. Ask the user to enter their ten quizzes, one midterm and one final exam grades
2. Drop the first or last quiz that has the lowest mark
3. Calculate the average of the nine remaining quizzes
4. Calculate the weighted average of each assessment
5. Calculate and output the final grade

Code:

```
#include <stdio.h>

double calculateFinalGrade(double midterm, double final_exam, double
quiz_avg){
```

```

    double quiz_marks=quiz_avg*0.25; //quiz constitutes 25% to the
final grade
    double midterm_marks,final_exam_marks, final_grade;
    if(midterm>=final_exam){
        midterm_marks=(midterm)*(0.35);
        final_exam_marks=(final_exam)*(0.40);
    }
    else{
        midterm_marks=(midterm)*(0.25);
        final_exam_marks=(final_exam)*(0.50);
    }

    final_grade = quiz_marks+midterm_marks+final_exam_marks;
    return(final_grade);

}

int main(void) {
    // Declare variables
    double quiz1, quiz2, quiz3, quiz4, quiz5, quiz6, quiz7, quiz8,
quiz9, quiz10;//variables for quizzes
    double midterm, final_exam, quiz_avg, final_grade, sum_quiz;

    // User input quiz marks
    printf("Enter 10 Quiz grades (on a scale of 0 to 10):\n");
    scanf("%lf %lf %lf %lf %lf %lf %lf %lf %lf %lf", &quiz1, &quiz2,
&quiz3, &quiz4, &quiz5, &quiz6, &quiz7, &quiz8, &quiz9, &quiz10);

    printf("Enter your midterm grade: ");
    scanf("%lf", &midterm);

    printf("Enter your final exam grade: ");
    scanf("%lf", &final_exam);

    // drop the lowest of first or last quiz
    if(quiz1<=quiz10){//if quiz 1 and quiz 10 are equal, drop quiz 1
        quiz1=0;
    }
}

```

```

else{
    if(quiz10<quiz1){
        quiz10=0;
    }
}

// calculate quiz average
sum_quiz = quiz1 + quiz2 + quiz3 + quiz4 + quiz5 + quiz6 + quiz7 +
quiz8 + quiz9 + quiz10;
quiz_avg = sum_quiz / 9.0;

final_grade = calculateFinalGrade(midterm,final_exam,quiz_avg);

printf("Your final grade is %.2lf.", final_grade);//final grade
outputted, rounded to two decimal places

return (0);
}

```

Output:

Part (a):

Enter 10 Quiz grades (on a scale of 0 to 10):

9.5 6 4 10 7.8 3.4 9 5.6 9 10

Enter your midterm grade: 73

Enter your final exam grade: 84

Your final grade is 62.05 percent.

Part (b):

Enter 10 Quiz grades (on a scale of 0 to 10):

9.5 8.4 9 10 7.8 10 9 9.6 9 10

Enter your midterm grade: 89

Enter your final exam grade: 81

Your final grade is 65.85 percent.

Part (c):

Enter 10 Quiz grades (on a scale of 0 to 10):

8.5 8.5 9 8.5 7.5 7 9 9.5 10 10

Enter your midterm grade: 80

Enter your final exam grade: 70

Your final grade is 58.19 percent.

Screenshot:

Part (a):

```
Output Clear
Enter 10 Quiz grades (on a scale of 0 to 10):
9.5 6 4 10 7.8 3.4 9 5.6 9 10
Enter your midterm grade: 73
Enter your final exam grade: 84
Your final grade is 62.05 percent.

=== Code Execution Successful ===
```

Part (b):

```
Output Clear
Enter 10 Quiz grades (on a scale of 0 to 10):
9.5 8.4 9 10 7.8 10 9 9.6 9 10
Enter your midterm grade: 89
Enter your final exam grade: 81
Your final grade is 65.85 percent.

=== Code Execution Successful ===
```

Part (c):

```
Output Clear
Enter 10 Quiz grades (on a scale of 0 to 10):
8.5 8.5 9 8.5 7.5 7 9 9.5 10 10
Enter your midterm grade: 80
Enter your final exam grade: 70
Your final grade is 58.19 percent.

=== Code Execution Successful ===
```

Problem #4:

Display a random card from a deck of cards with its suit.

Algorithm:

1. Output a random card type and a random suit from a deck of card

Code:

```
#include <stdio.h>
#include <time.h>
#include <stdlib.h>
#include <wchar.h>
#include <locale.h>

int main(void) {
    int rank, suit;
    setlocale(LC_CTYPE, "");

    wchar_t diamond = 0x2666; // daimond
    wchar_t club = 0x2663; // club
    wchar_t heart = 0x2665; // heart
    wchar_t spade = 0x2660; // spade

    srand(time(NULL)); // each time the programs runs, a different
    random number is generated
    rank = rand() % 13 + 1; // random number generated from 1 to 13 and
    stored in variable rank
    suit = rand() % 4 + 1; // random number generated between 1 to 4 and
    stored in variable suit

    // outputs random card type
    if (rank == 1) {
        wprintf(L"A");
    }
    else if (rank >= 2 && rank <= 10) {
        wprintf(L"%d", rank);
    }
    else if (rank == 11) {
        wprintf(L"J");
    }
}
```

```

    }
    else if (rank == 12) {
        wprintf(L"Q");
    }
    else if (rank == 13) {
        wprintf(L"K");
    }

    //outputs random suit type
    switch (suit) {
        case 1: wprintf(L"%lc\n", diamond); //daimond
        break;
        case 2: wprintf(L"%lc\n", club); //club
        break;
        case 3: wprintf(L"%lc\n", heart); //heart
        break;
        case 4: wprintf(L"%lc\n", spade); //spade
        break;
    }
    return (0);
}

```

Output:

4♣

Screenshot:

